

Program : Diploma in Instrumentation Engineering	
Course Code : 4084	Course Title: Analytical and Biomedical Instruments
Semester : 4	Credits: No Credits
Course Category: Program Core	
Periods per week: 3 (L: 3 T: 0 P: 0)	Periods per semester: 45

Course Objectives:

- To describe the methods and instrumentation systems used in analysis.
- To familiarize blood pressure measurements.
- To explain electric impulses in different parts of human body and familiarize with ECG, EMG and EEG.

Course Prerequisites:

Topic	Course code	Course name	Semester
Photometry and spectrometer		Applied physics II	2
Chromatography		Applied chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the spectrometric analysis using different types of photometers and spectrometers.	11	Understanding
CO2	Explain different types of chromatography.	10	Understanding
CO3	Understand principles of operations and construction details of different types of industrial gas analysers.	11	Understanding
CO4	Understand blood pressure measurement, ECG EMG and EEG.	11	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	2					
CO2	2	2					
CO3	2	2					
CO4	2	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the analysis of various samples using different types of photometers and spectrometers.		
M1.01	Explain the various regions of electromagnetic spectrum and types of spectrum (absorption and emission)	1	Understanding
M1.02	State the fundamental laws of photometry	1	Understanding
M1.03	Explain the basic components of photometers	1	Understanding
M1.04	Compare different types of spectroscopy (UV, Visible & IR)	2	Understanding
M1.05	Illustrate the construction and working of single beam and double beam filter photometer and spectrophotometer.	1	Understanding
M1.06	Explain the construction and working of flame photometers	1	Understanding
M1.07	Illustrate basic principles of Mass Spectroscopy	1	Understanding
M1.08	Outline the basic components of Mass spectrometer	1	Understanding
M1.09	Explain NMR Spectrometers	1	Understanding
M1.10	Explain Raman spectrometers and its applications	1	Understanding

Contents:

Electromagnetic spectrum – types of spectrum - Absorption spectrometers -fundamental laws of absorption photometry – classification of photometer & spectrometers, visible, UV, IR spectrometry -components of photometers and spectrometers - filter photometer – spectrophotometers – single beam and double beam - Emission Spectrometers - flame photometer Mass spectrometer– Types, Magnetic deflection mass spectrometer - Time of flight mass spectrometer, NMR Spectrometry, Basic principles, NMR spectrometer and Applications, Raman effect -Raman spectrophotometer, Applications

CO2	Explain different types of chromatography		
M2.01	Explain chromatography	2	Understanding
M2.02	Compare different chromatography Techniques such as Column Chromatography and Planar chromatography	2	Understanding
M2.03	Explain the working of paper chromatography and applications	2	Understanding
M2.04	Explain the working of gas chromatography	2	Understanding
M2.05	Explain the working of different types of liquid chromatography and applications	2	Understanding
	Series Test -1	1	

Contents:

Chromatography- Different techniques – Column chromatography - Planer Chromatography-Paper Chromatography -Applications .Gas chromatography – Sources-Detectors. Liquid chromatographs – sources- detectors- Applications – High-pressure liquid chromatographs – sources-detectors, Applications.

CO3	Explain principles of operations and construction details of different types of industrial gas analysers.		
M3.01	Illustrate the types of industrial gas analysers.	1	Understanding
M3.02	Explain the principle, construction and operation of Thermal conductivity gas analyzer.	1	Understanding
M3.03	Explain the principle, construction and operation of Paramagnetic oxygen analyzer(magnetic force type and wind type).	1	Understanding
M3.04	Explain the principle, construction and operation of Zirconia oxygen analyzer.	2	Understanding

M3.05	Explain the principle, construction and operation of Infrared analyzers(positive and negative filter type IR analyzer).	2	Understanding
M3.06	Explain the principle, construction and operation of Electrical conductivity analyzers.	2	Understanding
M3.07	Illustrate Applications of Gas analyzers.	2	Understanding
Contents: Industrial Analyzers -Thermal conductivity gas analyzer - Paramagnetic oxygen analyzer – magnetic force type and wind type - Zirconia oxygen analyzer - Infrared analyzers – positive and negative filter type IR analyzer- Electrical conductivity analyzer- Applications of Gas analysers			
CO4	Understand blood pressure measurement, ECG, EMG & EEG		
M4.01	Illustrate basic principle and working of direct type and indirect type BP measurement with diagram	1	Understanding
M4.02	Describe bio electric potential (resting and action potential)and sodium pump- transmission of impulses	1	Understanding
M4.03	Explain electrical activity of heart, ECG waveform , Einthoven’s triangle and different electrodes used for ECG measurement	2	Understanding
M4.04	Describe the recording method of ECG (Unipolar, bipolar and chest lead system) and the block diagram of ECG measurement	2	Understanding
M4.05	Explain EMG waveform and different electrodes used for EMG measurement with block diagram.	2	Understanding
M4.06	Explain Different frequency regions of EEG waveform and different electrodes used for EEG measurement(10– 20 electrode Lead system) with Block diagram.	3	Understanding
	Series Test – II	1	
Contents: Direct blood pressure measurement-Indirect blood pressure measurement –Bio electricity of a cell- Resting and action potentials of a cell- Sodium pump and transmission of impulses-- ECG wave form- Einthoven’s Triangle- Different electrodes used for ECG measurement- Unipolar lead - Bipolar lead and Chest lead system- Block diagram of an ECG machine- EMG waveform-Different electrodes used for EMG measurement- Block diagram of an EMG machine- Different frequency regions of EEG waveform- Different electrodes used for EEG measurement- The 10– 20 electrode Lead system- Block diagram of an EEG machine.			

Text / Reference:

T/R	Book Title / Author
R1	R.S. Khandpur, Handbook of Analytical Instruments- TMH.
R2	Chatwal – Anand, Instrumental methods of Chemical Analysis - Himalaya Publishing House.
R3	Liptak, B.G., Process Measurement and Analysis, CRC Press, 2005.
R4	G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004.
R5	Williard, Dean Merrit & Settle , Instrumental methods of analysis- CBS publishers.
R6	John H. Nelson, Nuclear Magnetic Resonance Spectroscopy, Prentice Hall/Pearson Education, 2003.
R7	Frank G. Kerry Industrial Gas Handbook: Gas Separation and Purification, Taylor and francis group, 2007.
R8	R. S. Kandpur,Hand Book of Biomedical Instrumentation -Tata MacGraw-Hill.
R9	Dr. Arumugham, Biomedical Instruments, Anuradha publications.
R10	Medical Instrumentation - John. G. Webster - Wiley India.
R11	Leslie Cromwell & Free. J. Weibell, Biomedical Instrumentation and Measurements- Prentice Hall.

Online Resources:

Sl.No	Website Link
1	http://www.sigmaaldrich.com/analytical-chromatography
2	http://www.slideshare.net/suniu/spectrophotometry-16091660
3	https://www.electrical4u.com/action-potential-and-resting-potential
4	https://en.wikipedia.org/wiki/Electrocardiography
5	http://www.slideshare.net/banuman35/applications-of-gas-chromatographyapplications-of-gc-by-pravisankar